# PayTrust: Relational Schema

## Relational schema

Notation: underlined = primary key, *italic* = foreign key, `?` = nullable. All ids are BIGINT UNSIGNED and all money is DECIMAL(15,2). Generated from the live database, so it matches the code exactly.

### Relations

**users** ( id, full_name, email, password_hash, role, kyc_status, is_active, created_at )

**wallets** ( id, *user_id*?, wallet_type, balance, created_at, updated_at )

**transactions** ( id, type, *from_wallet_id*, *to_wallet_id*, amount, idempotency_key, *initiated_by*?, note?, created_at )

**ledger_entries** ( id, *transaction_id*, *wallet_id*, entry_type, amount, balance_after, created_at )

**escrows** ( id, *buyer_id*, *seller_id*, amount, title, description?, status, *fund_txn_id*?, *settle_txn_id*?, created_at, funded_at?, completed_at?, settled_at?, updated_at )

**escrow_status_audit** ( id, *escrow_id*, old_status?, new_status, *changed_by*?, db_user, changed_at )

**escrow_create_requests** ( *buyer_id*, idempotency_key, *seller_id*, amount, title, *escrow_id*, created_at )

## Foreign keys

| Relation | Column | References |
|---|---|---|
| wallets | user_id | users(id) |
| transactions | to_wallet_id | wallets(id) |
| transactions | from_wallet_id | wallets(id) |
| transactions | initiated_by | users(id) |
| ledger_entries | wallet_id | wallets(id) |
| ledger_entries | transaction_id | transactions(id) |
| escrows | fund_txn_id | transactions(id) |
| escrows | buyer_id | users(id) |
| escrows | settle_txn_id | transactions(id) |
| escrows | seller_id | users(id) |
| escrow_status_audit | changed_by | users(id) |
| escrow_status_audit | escrow_id | escrows(id) |
| escrow_create_requests | escrow_id | escrows(id) |
| escrow_create_requests | buyer_id | users(id) |
| escrow_create_requests | seller_id | users(id) |

## Unique constraints

| Relation | Columns |
|---|---|
| users | email |
| wallets | user_id |
| transactions | idempotency_key |
| ledger_entries | transaction_id, entry_type |
| escrows | fund_txn_id |
| escrows | settle_txn_id |
| escrow_create_requests | escrow_id |

## Other constraints (enforced by MySQL)

| Relation | Rule |
|---|---|
| users | email must look like an email; full_name not blank |
| wallets | balance >= 0 (except the top-up system wallet); user wallets have a user_id, system wallets do not |
| transactions | amount > 0; from_wallet_id <> to_wallet_id |
| ledger_entries | amount > 0; one debit and one credit per transaction (UNIQUE) |
| escrows | amount > 0; buyer <> seller; any status past `created` needs fund_txn_id; `released` / `refunded` need settle_txn_id |
| escrow_status_audit | old_status <> new_status |

## View

**wallet_balance_summary** ( wallet_id, user_id, wallet_type, cached_balance, ledger_derived_balance, difference ), computed from wallets and ledger_entries.

## MongoDB collections (non-relational)

| Collection | Main fields |
|---|---|
| dispute_cases | escrow_id, raised_by, against, reason, description, status, evidence[ ], messages[ ], resolution { decision, resolved_by, resolved_at, note }, created_at, updated_at |
| kyc_documents | user_id, doc_type, number_masked, file_url, status, rejection_reason, submitted_at |
| activity_logs | user_id, action, entity_id, ip, metadata, created_at |